

Goal. Apply integration to calculate arc length, surface area, and fluid pressure.

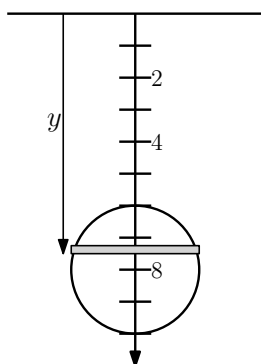
1. What is the formula for calculating the arc length of a curve? Describe the main idea behind how we derived this formula.
2. Show that the arc length of $f(x) = x^{3/2}$ from $x = 0$ to $x = 1$ equals $\frac{13\sqrt{13}-8}{27}$.
3. What is the formula for the surface area of a surface of revolution formed by rotating the graph $y = f(x)$ for $a \leq x \leq b$ about the x -axis? Describe the main idea behind how we derived this formula.

4. Find the surface area of the surface of revolution created by rotating the graph $y = x^3$ for $0 \leq x \leq 1$ about the x -axis.

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

5. A circular window on a submarine has a radius of 2 meters, and the center of the window is 8 meters below water level. Write and evaluate an integral to calculate the fluid force on the window by using the diagram below. Recall the density of water is 1000 kg/m^3 and the acceleration due to gravity is $g = 9.8 \text{ m/s}^2$.



Note it may be easier to evaluate the integral if you set up your axes with the point $(0,0)$ in the middle of the circle, as below. In this case, the depth no longer equals y ; you need to figure out an expression for the depth at y . Write another integral expressing the fluid force on the window by using the diagram below. Verify that these integrals are equivalent.

